# THE IMAGE OF THE SPECTRUM MAP OF A SPLIT ALGEBRA IS THE INTERSECTION OF THE CLOSED SETS CUT OUT BY THE EVALUATION IDEALS

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot, namely sources/stacks-chow/preamble.tex lines 14-18 (Xy-pic setup) and 154-217 (theorem declarations and mathematical macros). Nothing was added, removed or redefined, no package was added, and the excerpt serves only as preamble, never as a span coordinate.

*Editorial note.* Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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*Allowed standard prerequisites (editorial).*

- Spectrum of a quotient and the correspondence between ring maps out of it and points over the base. used at proof lines 4631-4633 when the image of Spec(S) to Spec(R) is compared with a closed subset. Identifies points of the image with evaluation maps that factor through S, which is what turns the claim into a statement about the ideals J_k.
- A prime contains an intersection if and only if it contains one of the factors. used together with the hint at proof line 4633. Converts the vanishing set of the intersection of the J_k into the union of the individual vanishing sets, which is the image.
- The description of the fibres of Spec(S) to Spec(R) over a point, in terms of the residue field of that point. implicit in the identification performed at proof lines 4631-4633. Supplies the evaluation maps Phi_k as exactly the points lying over the given prime, which is why their vanishing sets cover the image.

*Source reference.* Original lemma 0532: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6590–6590. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

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*Date*: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.2.** *Let $R$ be a ring. Let $S = R[T_1, \ldots, T_n]/J$. Assume $J$ contains elements of the form $P_i(T_i)$ with $P_i(T) = \prod_{j=1,\ldots,d_i}(T - \alpha_{ij})$ for some $\alpha_{ij} \in R$. For $\underline{k} = (k_1, \ldots, k_n)$ with $1 \leq k_i \leq d_i$ consider the ring map*

$$\Phi_{\underline{k}} : R[T_1, \ldots, T_n] \to R, \quad T_i \longmapsto \alpha_{ik_i}$$

*Set $J_{\underline{k}} = \Phi_{\underline{k}}(J)$. Then the image of $\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is equal to $V(\bigcap J_{\underline{k}})$.*

*Proof.* This lemma proves itself. Hint: $V(\bigcap J_{\underline{k}}) = \bigcup V(J_{\underline{k}})$. □